

Problem 1.4

April 19, 2026

1 AA 203 — Homework 1

1.1 Problem 1.4 — Zermelo's ship

1.1.1 a)

Minimum-time objective: $J = \int_{t_0}^{t_1} 1 \, dt = t_1 - t_0$, so the Hamiltonian is

$$H = 1 + p_x \left(v \cos \theta(t) + \frac{v}{h} y(t) \right) + p_y v \sin \theta(t)$$

State equations (dynamic constraints):

$$\dot{x}^*(t) = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} v \cos \theta^* + \frac{v}{h} y \\ v \sin \theta^* \end{bmatrix}$$

Costate (adjoint) equations:

$$\dot{p}^*(t) = \begin{bmatrix} -\frac{\partial H}{\partial x} \\ -\frac{\partial H}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 \\ -p_x \frac{v}{h} \end{bmatrix}$$

Since p_x is constant:

$$p_x = \text{const}, \quad p_y(t) = -p_x \frac{v}{h} t + \beta \quad \text{for some } \beta \in \mathbb{R}$$

Optimality condition (θ unconstrained):

$$\frac{\partial H}{\partial \theta} = 0 \quad \implies \quad 0 = -p_x v \sin \theta^*(t) + p_y v \cos \theta^*(t)$$

$$\implies \quad \tan \theta^*(t) = \frac{p_y(t)}{p_x} = \frac{-p_x \frac{v}{h} t + \beta}{p_x} = -\frac{v}{h} t + \frac{\beta}{p_x}$$

Setting $\alpha := \frac{\beta}{p_x} \in \mathbb{R}$:

$$\boxed{\tan \theta^*(t) = \alpha - \frac{v}{h} t}$$

Transversality condition (free terminal time):

Substituting $p_y(t_1^*) = -p_x \frac{v}{h} t_1^* + \beta$ (from the costate equation) into H , and using $y(t_1^*) = 0$:

$$\begin{aligned} H^*(t_1^*) &= 1 + p_x \left(v \cos \theta^*(t_1^*) + \frac{v}{h} \underbrace{y(t_1^*)}_{=0} \right) + \left(-p_x \frac{v}{h} t_1^* + \beta \right) v \sin \theta^*(t_1^*) \\ &= 1 + v \left[p_x \cos \theta^*(t_1^*) + \underbrace{\left(-p_x \frac{v}{h} t_1^* + \beta \right)}_{= p_x \tan \theta^*(t_1^*)} \sin \theta^*(t_1^*) \right] \\ &= 1 + v p_x \underbrace{\left(\cos \theta^* + \frac{\sin^2 \theta^*}{\cos \theta^*} \right)}_{= 1 / \cos \theta^*} \\ &= 1 + \frac{v p_x}{\cos \theta^*(t_1^*)} = 0 \\ &\implies p_x = -\frac{\cos \theta^*(t_1^*)}{v} \end{aligned}$$

1.1.2 b)

With $w(y(t)) \equiv \beta$ constant, the Hamiltonian becomes $H = 1 + p_x(v \cos \theta + \beta) + p_y v \sin \theta$, so

$$\dot{p}^*(t) = \begin{bmatrix} -\frac{\partial H}{\partial x} \\ \frac{\partial H}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies p_x^* \in \mathbb{R}, p_y^* \in \mathbb{R} \text{ (both constant)}$$

Optimality condition:

$$\frac{\partial H}{\partial \theta} = 0 \implies \tan \theta^*(t) = \frac{p_y^*}{p_x^*} = \text{const} \implies \theta^* = \text{const} := \theta_0$$

Transversality condition (free terminal time):

From the optimality condition $p_y = p_x \tan \theta_0$, substituting into H :

$$\begin{aligned} H^*(t_1^*) &= 1 + p_x(v \cos \theta_0 + \beta) + p_x \tan \theta_0 v \sin \theta_0 \\ &= 1 + p_x \left(v \cos \theta_0 + \beta + \frac{v \sin^2 \theta_0}{\cos \theta_0} \right) \\ &= 1 + p_x \cdot \frac{v + \beta \cos \theta_0}{\cos \theta_0} = 0 \end{aligned}$$

$$\Rightarrow p_x = -\frac{\cos \theta_0}{v + \beta \cos \theta_0}$$

Since θ_0 is constant, the trajectory integrates directly:

$$x(t) = (v \cos \theta_0 + \beta)(t - t_0) + x_0, \quad y(t) = v \sin \theta_0 (t - t_0) + y_0$$

Apply terminal BCs $x(t_1^*) = 0$, $y(t_1^*) = 0$. Let $T := t_1^* - t_0$:

$$(v \cos \theta_0 + \beta)T = -x_0 \quad (1)$$

$$v \sin \theta_0 T = -y_0 \quad (2)$$

Squaring and adding $(1)^2 + (2)^2$: $(vT)^2 = (x_0 + \beta T)^2 + y_0^2$, which expands to

$$(v^2 - \beta^2)T^2 - 2\beta x_0 T - (x_0^2 + y_0^2) = 0$$

Solving by the quadratic formula $T = \frac{2\beta x_0 \pm \sqrt{4[v^2 x_0^2 + (v^2 - \beta^2)y_0^2]}}{2(v^2 - \beta^2)}$:

Case (i) $v > \beta$: leading coefficient > 0 , discriminant $= 4[v^2 x_0^2 + (v^2 - \beta^2)y_0^2] > 0$ always. Taking the positive root:

$$t_1^* - t_0 = \frac{\beta x_0 + \sqrt{v^2 x_0^2 + (v^2 - \beta^2)y_0^2}}{v^2 - \beta^2}$$

No restriction on x_0 .

Case (ii) $v = \beta$: the T^2 term vanishes, leaving a linear equation $-2\beta x_0 T - (x_0^2 + y_0^2) = 0$:

$$t_1^* - t_0 = -\frac{x_0^2 + y_0^2}{2\beta x_0}$$

Requires $x_0 < 0$ for $T > 0$ (ship must start upstream of origin — it cannot make negative headway against the equal current).

Case (iii) $v < \beta$: leading coefficient < 0 ; the discriminant requires $v^2 x_0^2 \geq (\beta^2 - v^2)y_0^2$ for real solutions. Taking the smaller positive root:

$$t_1^* - t_0 = \frac{\beta x_0 + \sqrt{v^2 x_0^2 - (\beta^2 - v^2)y_0^2}}{v^2 - \beta^2}$$

Requires $x_0 < 0$ and $v^2 x_0^2 \geq (\beta^2 - v^2) y_0^2$ (the current is too strong to overcome; ship must start sufficiently upstream).

Please see plot.